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Abstract

As organizations increasingly rely on Artificial Intelligence (AI) to facilitate employee onboarding, the psychological impact of these tools remains underexplored. While AI offers promising support for knowledge acquisition, its potential to mitigate or exacerbate newcomer stress remains poorly understood. Drawing on Task-Technology Fit (TTF) theory and Self-Determination Theory (SDT), this study examines the relationship between AI-assistant task fit and perceived learning-related stress. A quantitative, cross-sectional survey ($N \approx 350$) targets newly hired employees in Germany. Data will be analyzed using parallel mediation analysis (PROCESS Model 4) to test direct and indirect effects. The study hypothesizes that high TTF is negatively associated with learning-related stress (H1). Furthermore, it is proposed that this relationship is mediated by the satisfaction of basic psychological needs: perceived autonomy (H2) and perceived competence (H3). The findings aim to demonstrate that AI effectiveness depends not merely on availability, but on its ability to scaffold employee independence and mastery. This offers practical guidance for HR practitioners designing psychologically supportive onboarding experiences in the digital age.

Keywords: Artificial Intelligence, Onboarding, Task-Technology Fit, Self-Determination Theory, Perceived Stress.

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¹ All students contributed equally to this seminar paper. As the paper was completed collaboratively, no individual parts are attributed to specific students. Therefore, we are fine with receiving a single, shared grade.

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List of abbreviations

AI	Artificial Intelligence
PSS	Perceived Stress Scale
SDT	Self-Determination Theory
TTF	Task-Technology Fit
W-BNS	Work-related Basic Need Satisfaction Scale

1 Introduction

Today, increasingly dynamic and knowledge-driven work environments require continuous learning (Mitschelen & Kauffeld, 2025), leading organizations to transform their workplaces into learning-oriented environments (Bucher et al., 2024). In this context, effective onboarding represents a crucial phase of organizational entry, supporting early knowledge acquisition and contributing to improved performance (Mitschelen & Kauffeld, 2025). At the same time, digitalization processes and the increasing use of Artificial Intelligence (AI) necessitate that employees adapt rapidly and autonomously to continuously evolving skill requirements. Generative AI, particularly conversational agents, offer a promising approach to supporting individualized learning in work environments (Bucher et al., 2024). To assess the current state of research, a structured literature review was conducted. Databases were systematically searched using predefined search strings across databases such as Scopus, Web of Science, PsycINFO and ScienceDirect. An overview of the literature search process is provided in Appendix A. The review revealed that existing research has primarily examined the motivational outcomes of AI-assisted learning, while the relationship between AI-assistant use and perceived stress remains largely underexplored (Wang & Wu, 2025). This gap is particularly relevant in onboarding contexts, since newly hired employees often face high uncertainty and a high cognitive load (Cheikh-Ammar et al., 2024). To address the identified research gap, this study investigates the relationship between the use of AI assistants and perceived stress during organizational onboarding. Based on this, the following research question was developed:

To what extent does the task-technology fit of AI assistants influence perceived learning-related stress during organizational onboarding and is this relationship mediated by perceived autonomy and perceived competence?

The following section outlines the theoretical framework and derives the hypothesis. Subsequently, the methodology is outlined and the chosen research method is justified. The paper concludes with a discussion of limitations, future research and practical implications.

2 Theoretical Background and Hypotheses Development

2.1 Task-Technology Fit and Stress

Beyond these aspects, the effectiveness of AI assistants in learning-intensive environments also depends on the Task-Technology Fit (TTF). Originally proposed by Goodhue and Thompson (1995) as a framework for evaluating the compatibility and suitability of a technology for improving performance, TTF posits that technological features must align with the specific requirements of the user's task. Furthermore, a high TTF is a prerequisite for efficiency, productivity and overall job satisfaction (Przegalinska et al., 2025). However, its role in mitigating negative psychological states, such as stress,

requires a specific conceptualization within the onboarding context.

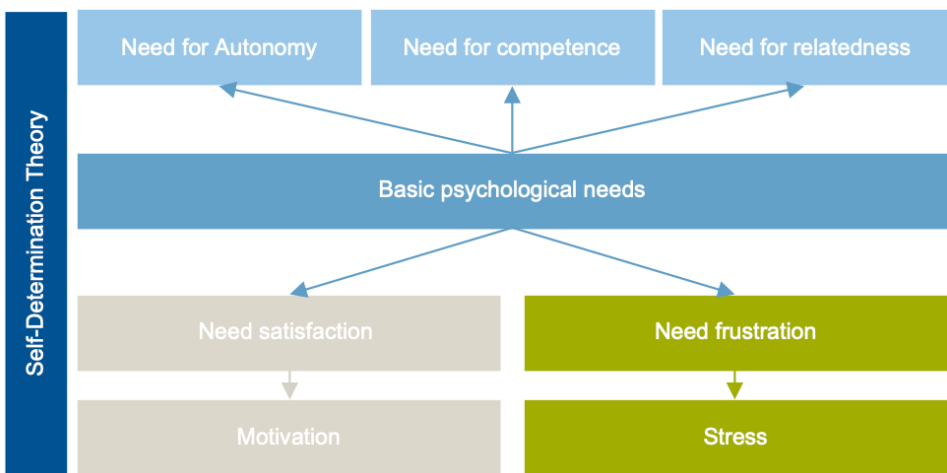
Perceived stress refers to an individual’s subjective appraisal of stress, which captures the extent to which their lives are experienced as uncontrollable and overloaded over a specific period of time (Klein et al., 2016). In this study, learning-related stress is focused on as a specific form of perceived stress. It refers to the extent to which work-related learning situations during the early phase of organizational entry are perceived as unpredictable and overloading. In such situations, newly hired employees must process large amounts of information and quickly understand new procedures and expectations, which can quickly become taxing. When AI assistants have a high degree of TTF, they can explain theoretical concepts step-by-step and answer questions at learners’ paces, making them feel more capable, less pressured and ultimately decreasing stress (Wang & Wu, 2025). On this basis, we hypothesize:

H1: *Higher task-technology fit of AI assistants is negatively associated with perceived learning-related stress during organizational onboarding.*

2.2 Self-Determination Theory

To distinguish the psychological processes underlying learning-related stress during onboarding for newly hired employees supported by AI assistants, the study draws on Self-Determination Theory (SDT). SDT posits that human motivation and engagement are driven by the satisfaction of three core psychological needs: *autonomy*, *competence* and *relatedness* (Wang & Wu, 2025), as illustrated by Figure 1. When these needs are satisfied, individuals experience higher intrinsic motivation, increased engagement and enhanced overall well-being. Conversely, when these needs are undermined, stress and pressure are more likely to occur, demonstrating a phenomenon known as need frustration (Olafsen et al., 2021).

Figure 1 *Self-Determination Theory Framework*



Note. Adapted from “Building a triadic model of technology, motivation and engagement: A mixed-

methods study of AI teaching assistants in design theory education. *Frontiers in Psychology* “, by Wang & Wu (2025).

Although SDT also emphasizes relatedness, this study focuses solely on autonomy and competence, as AI assistants primarily support individual learning and task-related competencies rather than social interaction.

2.3 The Mediating Role of Autonomy

According to SDT, perceived autonomy refers to the experience of volition and the sense that one originates one's own actions (Olafsen et al., 2021). In learning contexts, autonomy is especially fostered when individuals can decide how, when and at what pace they engage with learning tasks. AI assistants can offer on-demand support and enable independent, flexible, self-paced learning. These characteristics support self-determined learning paths, which strengthen their perception of autonomy (Wang & Wu, 2025). A strong alignment between task requirements and AI assistance capabilities can foster perceived autonomy by enabling learners to independently regulate their learning processes, thereby reducing learning-related stress (Wang & Wu, 2025). Accordingly, perceived autonomy is expected to mediate the relationship between a high TTF of AI assistants and learning-related stress, providing the basis for the following hypothesis:

H2: *Perceived autonomy mediates the relationship between AI task-technology fit and perceived learning-related stress.*

2.4 The Mediating Role of Competence

Perceived competence refers to the feeling of being effective and capable of successfully meeting situational demands (Deci & Ryan, 2000). When the capabilities of AI assistance are closely aligned with task requirements, high TTF enables AI assistants to support effective task completion, thereby reinforcing perceived competence. Through immediate feedback and clear guidance, AI assistants with a high TTF can reinforce learners' perceived competence (Wang & Wu, 2025). According to SDT, higher perceived competence is associated with lower need frustration and reduced learning-related stress (Olafsen et al., 2021). Accordingly, perceived competence is expected to function as a psychological mechanism through which AI TTF influences learning related stress, leading to the following hypothesis.

H3: *Perceived competence mediates the relationship between AI task-technology fit and perceived learning-related stress.*

3 Methodology Justification

To investigate the hypothesized relationships, a quantitative, cross-sectional survey design was chosen. This approach is justified on three theoretical and practical grounds. First, the core variables, perceived stress, autonomy and competence, are subjective psychological states. According to Spector (2006), self-report measures are the most appropriate method for assessing internal states that are not externally observable. Second, the study aims to test a specific mediation model involving indirect effects. Quantitative regression-based mediation analysis allows the statistical isolation of these mechanisms to test specific hypotheses (Hayes, 2018). Third, the target population consists of employees across various organizations. An anonymous online survey minimizes social desirability bias, the tendency for respondents to underreport negative states like stress or incompetence to appear favorable to their employer (Podsakoff et al., 2003).

3.1 Research Design

The study employs a cross-sectional online survey design. Data is collected via SoSci Survey, a professional web-based platform selected for its compliance with General Data Protection Regulation standards (Leiner, 2024). To minimize coverage error and ensure a diverse sample, a multi-channel recruitment strategy is utilized. The survey is distributed via professional and organizational channels, including LinkedIn, as it enables targeted access to early-career professionals, newly hired employees and working students across the DACH² region. Additionally, distribution via Technische Universität München (TUM) alumni communication channels target graduates which have recently entered the corporate world. The coverage of organizational channels can be extended to include students at the Duale Hochschule Baden Württemberg (DHBW), as they are currently employed and have likely been recently onboarded. Finally, to capture working students who may be less active on professional networks, the research team utilizes institutional (e.g. TUM Moodle & mailing lists) and social channels (e.g. TUM WhatsApp Groups) and conducts in-person recruitment at Bildungscampus Heilbronn.

The process of data collection process follows strict protocol to ensure quality. Prior to the main study, a pretest with a smaller sample population is conducted to evaluate item and instruction clarity, survey flow and duration. Based on feedback, adjustments can be made to ensure high-quality questionnaires and mitigate issues related to reliability and validity (Draugalis et al., 2008). Additionally, a reminder message can be sent after one week to improve response rates, consistent with recommended follow-up procedures (Draugalis et al., 2008).

3.2 Sample Description

The target population consists of employees and working students within German organizations who

² Germany (D), Austria (A), Switzerland (CH)

have been in their roles for six to twelve months. This onboarding period is characterized by high learning demands and uncertainty, making it ideal for examining perceived stress (Bauer et al., 2025; Woodrow & Guest, 2020). By centering the study on German organizations, the research accounts for a consistent working culture and regulatory environment (European Commission, 2020). To further ensure generalizability and statistical power, the study spans across diverse industries and roles, reflecting the cross-functional adoption of AI-based assistants (Soulami et al., 2024). This diversity, coupled with a deliberate focus on balanced gender and age distributions, helps address known demographic variances in stress responses (Heub et al., 2024). To tie this together, the use of stratified random sampling guarantees representativeness across industry, role, gender and age (Makwana et al., 2023).

Sample size planning followed a power-based approach appropriate for mediation analysis. Because indirect effects are harder to detect than direct ones, standard power calculations (e.g. G*Power) often underestimate the required sample size. Methodological recommendations for mediation analysis suggest that several hundred observations are necessary for robust estimation via bootstrapping (Fritz & MacKinnon, 2007; Hayes, 2018; Qin, 2023).

Following empirical studies of a similar nature in the field of AI adoption (Sohn & Kwon, 2020) and a purposive sampling strategy, the target sample size is set at 300 to 400 respondents. This range satisfies the requirements for structural equation modeling to achieve high statistical power (Hair et al., 2019). This volume also accounts for potential exclusions, missing data or power considerations and allows for adequately powered subgroup-specific analyses. Given that Germany recorded approximately 4.356 million new hires in 2024, this target is both feasible and representative of the substantial population undergoing employment transitions (Gürtzgen, 2025; Statistisches Bundesamt, 2025).

3.3 Survey Development

The survey is administered online using SoSci Survey. To minimize order effects and priming, the survey follows a funnel approach aligned with the hypothesized causal model.

Participants first complete the screening questions to confirm eligibility. The instrument then presents measures for the independent variable (TTF), followed by the mediating variables (Autonomy and Competence) and finally the dependent variable (Learning-Related Stress).

To mitigate common method bias, distinct section breaks are used (Podsakoff et al., 2003). Furthermore, an attention-check item was embedded in the questionnaire to identify and exclude careless respondents (Oppenheimer et al., 2009). Demographic and control variables (*Age, Gender and Industry*) were placed at the conclusion of the survey to prevent stereotype threat effects (Steele & Aronson, 1995).

The survey operationalizes the study's core constructs, using established, psychometrically sound scales. All items, except for the perceived stress scale, were measured on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree). This format was chosen to offer respondents more specific

choices and to maximize variance for statistical analysis, as 7-point scales have been shown to perform better than 5-point scales in this regard (Joshi et al., 2015).

The independent variable, TTF, was measured using an adapted version of the instrument developed by Goodhue and Thompson (1995). The original framework conceptualizes "fit" as a multi-dimensional construct comprising eight distinct factors: (1) *Quality* (currency and right level of detail), (2) *Locatability* (ease of finding data), (3) *Authorization* (access rights), (4) *Compatibility* (consistency across systems), (5) *Production Timeliness* (system speed), (6) *Systems Reliability* (uptime), (7) *Ease of Use/Training* and (8) *Relationship with Users* (IS support) (Goodhue & Thompson, 1995).

However, given the specific context of generative AI assistants used during organizational socialization, a parsimonious selection of dimensions was necessary to ensure content validity. Dimensions such as Authorization, Systems Reliability and Relationship with Users were excluded, as modern AI assistants are typically direct-access cloud platforms where server uptime is high and interaction with an internal IT support department is negligible. Furthermore, Locatability was deemed less relevant, as generative AI synthesizes answers rather than requiring users to navigate complex file structures to locate data. Consequently, this study focuses on the dimensions most critical to the learning and adaptation process:

1. **Quality (Precision):** Assessing if the AI provides the precise level of information needed for role clarity.
2. **Compatibility:** Assessing if the AI integrates smoothly with other work processes the new hire must learn.
3. **Production Timeliness:** Assessing if the AI accelerates the completion of onboarding tasks.
4. **Global Fit:** A summary measure capturing the overall alignment between tool functionality and task requirements.

The Items were contextually adapted to specifically reference "AI assistance" and "onboarding tasks" (see Appendix B).

Mediating psychological mechanisms were assessed using the validated subscales Work-related Basic Need Satisfaction Scale (W-BNS) from Van den Broeck et al. (2010). This scale was chosen because it is specifically designed to measure SDT needs in organizational contexts. The study assesses whether the AI assistant supports users' independent rather than imposing control, reflecting the need for autonomy.

Competence determines if the AI assistant enhances the user's feeling of capability during the learning process.

The dependent variable was measured using the full 14-item version of the Perceived Stress Scale (PSS-14) developed by Cohen et al. (1983). Participants rated the frequency of their feelings over the last month on a 5-point Likert scale (0 = Never to 4 = Very Often). The scale includes 14 items, including 7 positive framed and reverse coded. The final score is calculated as a summation, where a higher score

indicates a higher level of perceived learning-related stress. While the PSS measures global stress appraisals, this study aims to isolate domain-specific stress related to organizational entry. To ensure the instrument captured the specific "Demand-Resource" gap of the onboarding phase, the items were contextualized. General references to life were replaced with context-specific terms such as "...in your new job" or "...regarding your onboarding process." This ensures that the reported stress levels are attributable to the work environment and the learning curve, rather than extraneous private life factors.

Sample Item: *In the last month, how often have you been upset because of something that happened unexpectedly in your new job?*

3.4 Planned Data Analysis

Analyses will be conducted in SPSS (including the PROCESS macro) and will include descriptive statistics, reliability checks, regression and regression-based mediation with bootstrapped confidence intervals (Field, 2013; Hayes, 2018).

Missing data will be assessed; if minimal, listwise deletion may be used and if substantial, an appropriate method will be documented (Field, 2013). Items will be coded consistently (including reverse scoring where required) and composite scores will be computed for TTF, autonomy, competence and learning-related stress. Composite scores will be computed as scale means, using the original response formats of the instruments to preserve their validated measurement properties. Specifically, TTF and the W-BNS autonomy and competence subscales are rated on a 7-point Likert scale, whereas perceived learning-related stress is measured with an adapted PSS-10 using a 1-5 frequency response format; negatively worded items will be reverse-coded before aggregation (Cohen et al., 1983; Van den Broeck et al., 2010).

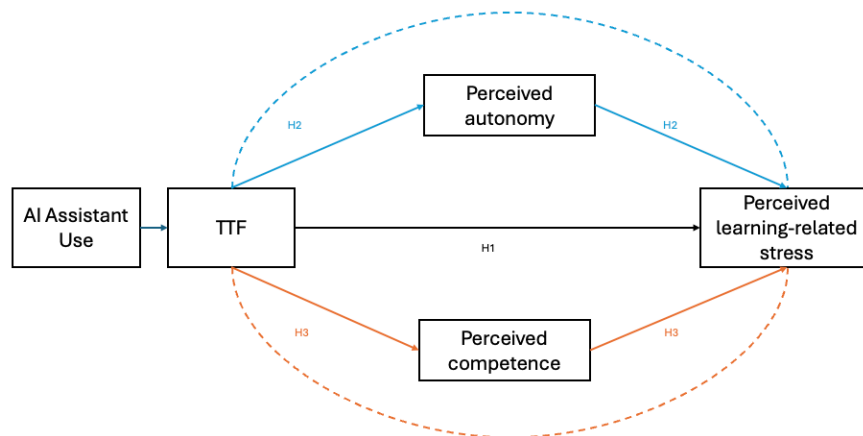
Scale reliability will be evaluated using Cronbach's alpha, with values around 0.70 or higher considered acceptable for research purposes (Nunnally, 1978). Descriptive statistics (mean and standard deviation) will be reported. Basic checks for unusual responses and influential observations will be conducted using standard diagnostics and handled conservatively to avoid removing meaningful variation (Tabachnick & Fidell, 2019).

These methods were selected because the study uses continuous scale variables from a cross-sectional survey and tests both a direct relationship (TTF -> stress) and theory-driven indirect mechanisms (via autonomy and competence). Regression provides a clear test of the direct association, while mediation with bootstrapped confidence intervals is recommended for indirect effects because these effects often have non-normal distributions (Field, 2013; Preacher & Hayes, 2004; Hayes, 2018)

To test **H1**, a linear regression model will be estimated because it is an appropriate approach for assessing the relationship between a continuous predictor (TTF) and a continuous outcome (learning-related stress) in cross-sectional survey data (Field, 2013).

To test **H2** and **H3**, mediation will be estimated using a regression-based approach (e.g., PROCESS Model 4), which is suitable for testing indirect effects via autonomy and competence. Indirect effects will be estimated using bootstrapping (5000 resamples) because indirect effects often have non-normal sampling distributions and bootstrapping provides robust confidence intervals (Preacher & Hayes, 2004; Hayes, 2018). Mediation is supported when the 95% bootstrap confidence interval excludes zero (Hayes, 2018). Each mediator will be tested separately, followed by a parallel mediation model including autonomy and competence simultaneously to examine their unique contributions (Hayes, 2018).

Figure 2 *Conceptual research model and hypotheses*



Note. Own work

Results will be reported following APA 7, including coefficients, p-values and confidence intervals for direct and indirect effects, with mediation interpreted primarily via the bootstrapped indirect effect confidence interval (Hayes, 2018). All analytic decisions (exclusions, scoring rules and handling of missing data) will be documented. Given cross-sectional self-report data, limitations regarding causal inference and common method bias will be acknowledged; anonymity and established measures reduce these risks and will be described (Podsakoff et al., 2011). A codebook and analysis syntax will be stored and can be shared upon request to support replication.

4 Discussion

4.1 Expected findings

It is expected that a higher TTF of AI assistants will be associated with lower perceived learning-related stress during organizational onboarding. Because onboarding is a learning-intensive and often uncertain process, AI tools are expected to reduce cognitive overload by providing timely explanations and structured guidance (Przegalinska et al., 2025). Therefore, support for H1 is expected as a negative

relationship between TTF and learning-related stress.

It is furthermore expected that this relationship is partly explained by perceived autonomy and perceived competence, consistent with SDT. When AI support aligns with task requirements, employees may feel more in control of their learning pace (autonomy) and more capable of mastering new tasks (competence), which should reduce stress (Olafsen et al., 2021). Accordingly, significant indirect effects for both mediators are expected, supporting H2 and H3 (Hayes, 2018).

4.2 Conceptual and practical contributions

This study contributes to the literature on business psychology and AI-at-work research by addressing perceived learning-related stress during organizational onboarding, rather than only performance or motivation outcomes. By integrating TTF with SDT, the model clarifies when AI assistants should be helpful in onboarding and why they may reduce stress. The proposed model specifies autonomy and competence as key mechanisms linking TTF to lower stress, extending SDT-based well-being research to AI-supported onboarding contexts (Deci & Ryan, 2000; Olafsen et al., 2021). It also extends TTF work by connecting fit to well-being outcomes (Goodhue & Thompson, 1995). Practically, the findings can guide Human Resources departments and Learning & Development teams to evaluate AI assistants based on task fit, not just availability or technical features. The model highlights autonomy and competence as central design targets to reduce stress and improve onboarding experiences. Building on these insights, the model further offers actionable implementation checks by monitoring perceived fit, autonomy, competence, and stress to identify where AI support is helpful versus overwhelming (Hayes, 2018).

4.3 Limitations

Despite the robustness of the design, several limitations of the intended study should be acknowledged. First, a cross-sectional design limits causal conclusions. Although mediation analyses allow the examination of indirect relationships, the temporal ordering of AI assistant use, psychological need satisfaction and perceived stress cannot be definitively established. Future research would benefit from longitudinal designs tracking newcomers over the first months. Second, the reliance on self-report instruments may introduce common method bias, social desirability bias, or inaccurate self-assessment. However, perceived stress, autonomy and competence represent subjective experiences that are most appropriately captured via self-reports. The use of validated scales and anonymous data collection may mitigate but not fully eliminate those concerns. This is particularly relevant in onboarding contexts, where newly hired employees may hesitate to report stress, despite anonymity.

Moreover, the focus on newly hired employees in German organizations may limit the generalizability of the findings to other cultural or organizational contexts. However, this focus was chosen to ensure contextual consistency regarding digital infrastructure, labor regulations and emerging AI governance

frameworks. Finally, the study captures a snapshot within a fast-evolving technological environment. As AI assistants continue to develop, patterns of use and their psychological effects may change. Hence, the findings should be interpreted as a snapshot of early adoption rather than as stable long-term effects.

4.4 Implications for business psychology contexts

The anticipated findings have important implications for business psychology, particularly in the context of employee onboarding and learning. The results offer a blueprint for “psychologically safe” digitization in onboarding. Human Resources and IT departments should evaluate AI assistants not solely on technical features and availability, but on Task Fit. A tool that is technically advanced but ill-fitted to the specific context of onboarding may increase stress rather than reduce it. To minimize stress, onboarding programs should be designed to maximize the users’ experience of autonomy. For example, organizations should encourage self-paced interaction with AI tools rather than mandating rigid usage protocols. To support the experience of competence, AI implementations should be configured to provide transparent and constructive feedback that reinforces the user’s ability to solve problems independently, rather than black-box answers that foster dependency. This is particularly important in remote or hybrid work environments, where newly hired employees often rely heavily on digital tools to structure their work and learning. From a business psychology perspective, a psychologically informed integration of AI assistants may not only reduce stress but also support employee well-being, engagement, and long-term effectiveness.

As organizations increasingly rely on AI to scale their onboarding processes, understanding the human side of this technology is crucial. This study proposes that the efficacy of AI assistants is not intrinsic to the technology itself but is contingent on its fit with the users’ tasks and psychological needs. By demonstrating that high TTF can mitigate learning-related stress through the enhancement of autonomy and competence, this research highlights the potential of AI to serve as a supportive, rather than disruptive force in the future of work.

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Appendix A

Extract of Literature Review

Year	Author	Journal	Title
2025	Ahmed A. Aldraiweesh and Uthman Alturki	IEEE Access	The Influence of Social Support Theory on AI Acceptance: Examining Educational Support and Perceived Usefulness Using SEM Analysis
2025	Annabell Mitschelen, Simone Kauffeld	Frontiers in Organizational Psychology	Workplace learning during organizational onboarding: integrating formal, informal, and self-regulated workplace learning
2025	Awais Ahmed and Hugo Terashima-Mari	Journal of Computers in Education	Adoption of AI-coding assistants in programming education: exploring trust and learning motivation through an extended technology acceptance model
2025	Bai, Y., Wang, S.	Scientific Reports	Impact of generative AI interaction and output quality on university students' learning outcomes: a technology-mediated and motivation-driven approach
2024	Bucher, Andreas & Schenk, Birgit & Dolata, Mateusz & Schwabe, Gerhard	LCIS 2024 Conference Proceedings	WHEN GENERATIVE AI MEETS WORKPLACE LEARNING – CREATING A REALISTIC & MOTIVATING LEARNINGEXPERIENCE WITH A GENERATIVE PCA
2025	C. Cubillos, R. Mellado, D. Cabrera-Paniagua, E. Urria	IEEE Access	Generative Artificial Intelligence in Computer Programming: Does It Enhance Learning, Motivation, and the Learning Environment?
2024	Chen, C., Hu, W., & Wei, X.	Interactive Learning Environments	From anxiety to action: exploring the impact of artificial intelligence anxiety and artificial intelligence self-efficacy on motivated learning of undergraduate students.
2025	Dr. Syed Azhar Hussain, Saher Shafiq, Syed Anwar Ali, Muhammad Azam	ACADEMIA International Journal for Social Sciences	The Impact of AI Powered Learning Platforms on Student Motivation and Academic Self-Efficacy: A Psychological Perspective
2021	Fernández Salasero, José, Román-Graván, Pedro, Reyes-Robelo, Miguel-María, Montero-Rueda, María	International Journal of Environmental Research and Public Health	Impact of Educational Technology on Teacher Stress and Anxiety: A Literature Review
2024	George Hanshaw	Scientific Reports	Exploring the Effectiveness of AI Course Assistants on the Student Learning Experience
2025	Ghada ElSayad, Hebat Allah Mamdouh	Journal of Applied Research in Higher Education	AI-driven learning: evaluating the role of chatbots in enhancing student motivation and learning outcomes
2025	Hong, X., Guo, L.	Education and Information Technologies	Effects of AI-enhanced multi-display language teaching systems on learning motivation, cognitive load management, and learner autonomy
2025	Huang, W., Jiang, J., King, R.B. et al.	International Journal of Educational Technology in Higher Education	Chatbot and student motivation: a scoping review
2025	Huili Yang, Yanping Rui	Acta Psychologica	Transforming EFL students' engagement: How AI-enhanced environments bridge emotional health challenges like depression and anxiety
2024	Jan Henrik Gruenhagen et al.	Computers and Education: Artificial Intelligence	The rapid rise of generative AI and its implications for academic integrity: Students' perceptions and use of chatbots for assistance with assessments
2021	Kaat Iterbeke, Kristof De Witte, Wouter Schelfhout	Computers in Human Behavior	The effects of computer-assisted adaptive instruction and elaborated feedback on learning outcomes. A randomized control trial
2024	Qiwei Zhou, Keyu Chen, Shuang Cheng	Technological Forecasting & Social Change	Bringing employee learning to AI stress research: A moderated mediation model
2025	Richmond Kakra Addae, Cecilia Brown	IJRIS	The Role of Artificial Intelligence in Enhancing Human Longevity: Mitigating Cognitive Overload for Extended Lifespan Among Master's Students in Mathematics at the Catholic University of Ghana
2009	Saadé, Raafat; Kira, Dennis	Journal of Information Technology Education	Computer Anxiety in E-Learning: The Effect of Computer Self-Efficacy
2025	Saeed Awadh Bin-Nashwan	Foresight	ChatGPT and academic integrity: the role of personal best goals, academic competence, and workplace stress
2024	Samuel BoguslawskiCorresponding Author: Rowan Dear, Mark G. Dawson	Information and Learning Sciences	Programming education and learner motivation in the age of generative AI: student and educator perspectives
2024	Shahzad, M.F., Xu, S. & Zahid, H	Education and Information Technologies	Exploring the impact of generative AI-based technologies on learning performance through self-efficacy, fairness & ethics, creativity, and trust in higher education
2023	Thomas K.F. Chiu	Interactive Learning Environments	Teacher support and student motivation to learn with Artificial Intelligence (AI) based chatbot
2025	Tummalapenta, S.R., Pasupuleti, R.S., Chebolu, R.M. et al.	Journal of Computers in Education	Factors driving ChatGPT continuance intention among higher education students: integrating motivation, social dynamics, and technology adoption
2022	Wang, Y., Wei, C., Lin, H., Wang, S.	Interactive Learning Environments	What drives students' AI learning behavior: a perspective of AI anxiety
2025	Weixuan Huang, Xinyi Ding, Shijia Shao	Environment and Social Psychology	A study on the dual mechanisms of AI teaching assistants' impact on college students' learning motivation: From the perspective of social cognitive theory
2024	Woo, D.J., Wang, D., Guo, K. et al.	Education and Information Technologies	Teaching EFL students to write with ChatGPT: Students' motivation to learn, cognitive load, and satisfaction with the learning process
2025	Wu, CH., Ho, V.T.	Education and Information Technologies	Critical factors for why ChatGPT enhances learning engagement and outcomes
2022	Yahoudi, Fehus, Achouche, Riad, Cherqui, Walid, Brika, Saif, Abmecher, Mohamed, Mousa, Adam, Lemari, Hame	Frontiers in Psychology	The Impact of e-Learning Systems on Motivating Students and Enhancing Their Outcomes During COVID-19: A Mixed-Method Approach
2022	Yen-Fen Lee, Gwo-Jen Hwang & Pei-Ying Chen	Education Tech Research Dev	Impacts of an AI-based chabot on college students' after-class review, academic performance, self-efficacy, learning attitude, and motivation
2024	Yin, J., Goh, TT. & Hu, Y	International Journal of Educational Technology in Higher Education	Interactions with educational chatbots: the impact of induced emotions and students' learning motivation
2024	Yitong Fan, Lushan Tang, Huibao Lu, Huijie Shen, Shuang Tan, Yanying Zhao, Yuan Shen, Xinyu Li, Deyang Galsen	British Journal of Educational Technology	Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance
2025	Zhai, S., Zhang, S., Rong, Y., & Rong, G.	Educ. Inf. Technol.	Technology-driven support: exploring the impact of artificial intelligence on mental health in higher education
2025	Zijian Tang, Jun Liao	Acta Psychologica	Unlocking emotional resilience: Exploring the impact of AI-enhanced support systems on EFL teachers' burnout and EFL students' well-being in modern classrooms

Note. This section is only an exemplary excerpt of our literature review and does not claim to be exhaustive.

Appendix B

Survey



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Page 01

01

1. Introduction

ID01

Welcome to the Study on AI in Onboarding

What is this study about? We are investigating how digital tools, specifically AI assistants, support newly hired employees during their induction and learning phase.

Why participate?

Your insights are valuable. They will help us understand the psychological impact of AI in modern workplaces. This research is conducted as part of a Business Psychology project at the Technical University of Munich (TUM).

Data Protection & Anonymity:

- Your participation is strictly anonymous.
- No personally identifiable information (IP addresses, names, employer names) will be stored.
- The data will be used solely for academic research purposes.

Duration: The survey is concise and will take approximately 5–7 minutes to complete.

There are no right or wrong answers; we are strictly interested in your personal experience and honest opinion.

Please click "Next" to begin.

2. Employment Status

SC01

Please state your current employment status.

[Please choose] 

SC02

3. Onboarding Tenure

Please state the time of your onboarding tenure.

[Please choose] ▾

4. AI Usage

SC03

Please indicate whether AI was used during your onboarding.

- ☐ Yes
- ☐ No

5. Location

SC04

Are you currently employed in Germany?

- ☐ Yes
- ☐ No

Page 02

02

PHP code

```
// Check Employment: If they selected Option 4 (Not employed)
// Check Tenure: If they selected Option 3 (>12 months)
// Check AI Use: If they selected Option 2 (No)

if (
    (value('SC01') == 4) ||
    (value('SC02') == 3) ||
    (value('SC03') == 2)
) {
    // If ANY of the above are true, send them to the screenout page
    goToPage('Screenout');
}
```

TF02

6. Suitability of the AI Assistant

Please rate your experience with the AI assistant you used most frequently during your onboarding tasks.

	1 (Strongly Disagree)							7 (Strongly Agree)
The AI assistant provides the precise information I need to complete my onboarding tasks.	☐	☐	☐	☐	☐	☐	☐	☐
The AI assistant is compatible with the other work processes I have to learn.	☐	☐	☐	☐	☐	☐	☐	☐
The AI assistant helps me complete my learning tasks without unnecessary delays.	☐	☐	☐	☐	☐	☐	☐	☐
The functions of the AI assistant are a good fit for the learning requirements of my new job.	☐	☐	☐	☐	☐	☐	☐	☐
The AI assistant is missing critical functions that would be useful for my training.	☐	☐	☐	☐	☐	☐	☐	☐

SD01

7. Personal Choice in Using AI

Please rate your experience regarding your personal choice and freedom when using the AI assistant.

	1 (Strongly Disagree)	7 (Strongly Agree)
I felt free to choose how I used the AI assistant to solve problems.	☐	☐
I felt I had a choice in how I integrated the AI assistant into my workflow.	☐	☐
I felt like the decision to use the AI assistant for certain tasks was mine.	☐	☐
I felt forced to use the AI assistant in a specific way.	☐	☐
Please select "Strongly Disagree" for this statement.	☐	☐

SD02

	1 (Strongly Disagree)	7 (Strongly Agree)
I felt competent when using the AI assistant to complete my work.	☐	☐
I felt I could successfully master the new tasks with the help of the AI.	☐	☐
I felt confident in my ability to operate the AI assistant effectively.	☐	☐
Using the AI assistant made me feel capable in my new role.	☐	☐

LS01

8. Your Onboarding Experience

The following questions ask about your feelings and thoughts regarding your onboarding and learning process during the last month.

	1 (Never)				4 (Very Often)
In the last month, how often have you been upset because of something that happened unexpectedly in your new job?	☐	☐	☐	☐	☐
In the last month, how often have you felt that you were unable to control the important aspects of your onboarding process?	☐	☐	☐	☐	☐
In the last month, how often have you felt nervous and “stressed” about learning new work tasks?	☐	☐	☐	☐	☐
In the last month, how often have you dealt successfully with irritating work-related hassles?	☐	☐	☐	☐	☐
In the last month, how often have you felt that you were effectively coping with important changes that were occurring in your work environment?	☐	☐	☐	☐	☐
In the last month, how often have you felt confident about your ability to handle your work-related problems?	☐	☐	☐	☐	☐
In the last month, how often have you felt that things were going your way in your new role?	☐	☐	☐	☐	☐
In the last month, how often have you found that you could not cope with all the learning tasks that you had to do?	☐	☐	☐	☐	☐
In the last month, how often have you been able to control irritations in your daily work?	☐	☐	☐	☐	☐
In the last month, how often have you felt that you were on top of things regarding your new tasks?	☐	☐	☐	☐	☐
In the last month, how often have you been angered because of things that happened that were outside of your control during your training?	☐	☐	☐	☐	☐
In the last month, how often have you found yourself thinking about work tasks that you have to accomplish?	☐	☐	☐	☐	☐

In the last month, how often have you been able to control the way you spend your time learning?

☐☐☐☐☐

In the last month, how often have you felt learning difficulties were piling up so high that you could not overcome them?

☐☐☐☐☐

9. Age

DE01 

Please indicate your age

Age

Which gender do you identify with?

DE02 

- ☐ Female
- ☐ Male
- ☐ Non-binary / Diverse
- ☐ Prefer not to say

Which industry does your organization belong to?

DE03 

- ☐ Technology / Software / IT
- ☐ Finance / Banking / Insurance
- ☐ Consulting / Professional Services
- ☐ Engineering / Manufacturing / Automotive
- ☐ Healthcare / Pharma / Life Sciences
- ☐ Education / Research / Science
- ☐ Retail / Consumer Goods
- ☐ Public Sector / Government
- ☐ Media / Telecommunications
- ☐ Other

Thank you for your interest.

Unfortunately, you do not meet the specific inclusion criteria required for this particular study.

To ensure valid scientific results, our current research focus is strictly limited to:

Employees who are currently employed (including working students).

Individuals who joined their organization within the last 12 months.

Users who actively utilize AI assistants in their work.

We sincerely appreciate your willingness to support our research and hope you might participate in future studies.

You may now close this window.

Thank you for completing this questionnaire!

We would like to thank you very much for helping us.

Your answers were submitted, you may close the browser window or tab now.